AGNI-NETRA

AI Geospatial Network for Industrial Thermal Risk & Anomaly Analysis

INTELLIGENCE DOSSIER • Generated on 2026-09-12 07:15 UTC • DECISION SUPPORT SYSTEM

Event Code: EVT-GUJ-20260907-0AF4	**Classification:** Industrial Fire (97.1%)
Location: 22.35420°N, 69.86440°E (Gujarat)	**AGNI-NETRA Risk Level: CRITICAL** (80.3/100)
Active Detections: 6 observations	**Peak Radiative Power (FRP):** 285.0 MW
First Detected: 2026-09-07 11:54:19	**Last Detected:** 2026-09-07 11:54:19

1. Explainable AI (SHAP) Attribution

Evaluated via candidate XGBoost v3.0 model with TreeExplainer SHAP attribution.

Feature Name	Value	Shapley Impact	Direction
dist_to_facility_m	181.91402943653523	+0.420	Supports Class (+)

2. Geospatial Facility Association

Associated Facility: None (Uncataloged Site)	**Facility Type:** N/A
Facility Status: KNOWN	**Distance to Boundary:** 181.9 m
Land Cover Context: Industrial	**Source Provenance:** OSM / Satellite Survey

3. Risk Evaluation & Drivers

• Severe radiative heat output (Peak FRP: 285.0 MW)
• Current FRP (285.0 MW) is statistically critical (+6.4σ above historical baseline mean of 125.0 MW).
• Chronic persistent thermal source with continuous multiday emissions
• Direct correlation with high-hazard industrial facility boundary

4. Model Governance & Data Provenance

Model Version: XGBoost V3 (xgb-v3.0-real-candidate • Platt-Calibrated)	**Telemetry Source:** NASA FIRMS S-NPP/VIIRS 375m NRT
Spatial Context Engine: PostGIS 3.4 Multi-Layer Topology	**Verification Status:** ACTIVE (Pending Human Verification)
Automated Dispatch: DISABLED / GATED SAFE (Human Decision Support)	**Dossier Generated:** 2026-09-12 07:15:43 UTC